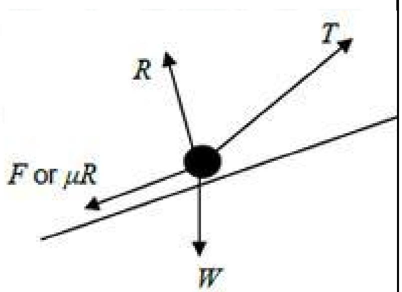
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Resolves horizontally and vertically to obtain expressions for F and R (allow consistent mixing of sin and cos)	AO3.4	M1	Resolving vertically $R = 8 \times 9.8 - 50\sin 40°$ Resolving horizontally $F = 50\cos 40°$
	Obtains correct expressions for R and F	AO1.1b	A1	
	States friction model $F = \mu R$ with 'their' values for F and R	AO1.2	B1	$F = \mu R$ $50\cos 40° = \mu(8 \times 9.8 - 50\sin 40°)$ $\mu = \dfrac{50\cos 40°}{8 \times 9.8 - 50\sin 40°}$
	Completes a rigorous argument that results with correct μ **AG** Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$\mu = 0.8279660445 = 0.83$ (2sf) **AG**
(b) (i)	Draws correct diagram with exactly four forces showing arrow heads and labels Can use Mg or $8g$ or 78.4 for W	AO3.3	B1	
(ii)	Resolves perpendicular to the plane resulting in a three term equation containing; R, $8g\cos 5°$ and $T\sin 40°$ (or $T\cos 50°$) OR Resolves horizontally and vertically to obtain equations of motion in horizontal and vertical	AO3.1b	M1	$R = 8 \times 9.8\cos 5° - T\sin 40°$ $= 78.1 - T\sin 40°$

directions			
Obtains correct expression for R OR Obtains correct horizontal and vertical equations	AO1.1b	A1	$T\cos 40° - 8 \times 9.8\sin 5° - F = 8 \times 3$ $T\cos 40° - 8 \times 9.8\sin 5° - 0.827\ldots \times (8 \times 9.8\cos 5° - T\sin 40°) = 24$
Forms a four term equation of motion parallel to the plane with correct terms (allow sign errors) OR Eliminates R to solve for T	AO3.1b	M1	$T = \dfrac{24 + 8 \times 9.8\sin 5° + 0.827\ldots \times 8 \times 9.8\cos 5°}{\cos 40° + 0.827\ldots \times \sin 40°}$ $T = 73.55939193 = 74$ *(2 sf)*
Obtains correct equation of motion OR Obtains correct expression(s) without R	AO1.1b	A1	
Uses the friction model in the four term equation of motion where R is in the form $a - bT$ and a and b are positive constants OR Uses the friction model in the horizontal and vertical equations Dependent on previous M1	AO3.1b	dM1	
Solves for T	AO1.1b	A1	
Obtains correct value of T with 2 sf accuracy. FT incorrect value found for T provided both M1 marks and dM1 mark have been awarded	AO3.2a	A1F	
			Total 12 marks

Q2.

(a) $\int x \, dx = \int 150\cos 2t \, dt$

Correct separation; condone missing $\int$ signs; must see dx, dt

B1

$\frac{1}{2}x^2 = 75\sin 2t \quad (+C)$

Correct integrals

Accept $\frac{1}{2} \times 150$

B1B1

$\left(20, \frac{\pi}{4}\right) \quad \frac{1}{2} \times 20^2 = 75\sin\left(2 \times \frac{\pi}{4}\right) + C$

C present. Use $\left(20, \frac{\pi}{4}\right)$ *to find C*

M1

$C = 125$

F on $x^2 = k \sin 2t$

A1F

$x^2 = 150 \sin 2t + 250$

Correct integrals and evaluation of C

A1 **6**

(b) (i) $t = 13 \quad x^2 = 150 \sin 26 + 250$ (= 364.38)

Evaluate $x^2 = f(13)$; $x^2 = k \sin 2t + c$ with numerical k and t

M1

$x = 19.1$ (cm)

AWRT

A1 **2**

(ii) $\left.\begin{array}{l} x = 11 \qquad \sin 2t = -\frac{129}{150} \quad (= -0.86) \\ \quad\text{or} \qquad 2t = -1.035\ldots,\ 4.176\ldots \end{array}\right\}$

M1

$t = 2.1$ (seconds)

AWRT

A1

2

[10]

Q3.

(a) $(V_H =)\ \frac{38.4}{2.4}$ = 16 m s^{-1}

M1: Horizontal range divided by time.
A1: Correct speed.

M1A1

2

(b) $3 = V_v \times 2.4 - \frac{1}{2} \times 9.8 \times 2.4^2$

M1: Equation to find the vertical component, with
s = ±3, t = 2.4 and a = ±g or ±9.8 or ±9.81.
A1: Correct equation with g or 9.8 or ±9.81.

M1A1

$V_v = \frac{3+28.224}{2.4}$ = 13.01

A1: Correct vertical component. Accept AWRT 13.

A1

$V = \sqrt{13.01^2 + 16^2}$ = 20.6 m s^{-1}

dM1: Finding speed using their answer from part (a) and their vertical component.
A1:Correct final speed. Accept AWRT 20.6

dM1A1

5

(c) $\tan\alpha = \frac{13.01}{16}$ or $\sin\alpha = \frac{13.01}{20.6}$ or $\cos\alpha = \frac{16}{20.6}$

M1: Trig equation to find the angle with:
cos with 13 or 16 in the numerator and 20.6 in denominator
sin with 13 or 16 in the numerator and 20.6 in denominator
tan with 13 and 16 in any position
A1F: Correct equation.

M1A1F

α = 39.1°

A1F: Correct angle. Accept AWRT 39°

A1F

Follow through incorrect answers to part (a) and (b), provided their speed from (b) is the resultant of two components.

3

[10]

Q4.

(a) $\mathbf{a} = \frac{dv}{dt}$

M1 for either term correct

M1

$= 4\pi\sin\left(\frac{\pi}{3}t\right)\mathbf{i} - 18t\,\mathbf{j}$

Accept $12 \times \frac{\pi}{3}\sin\left(\frac{\pi}{3}t\right)\mathbf{i} - 18t\,\mathbf{j}$

Condone no $\mathbf{i}$ *in (a)*

A1

2

(b) (i) Using $\mathbf{F} = m\mathbf{a}$:

$\mathbf{F} = 4 \times \left[-4\pi\sin\left(\frac{\pi}{3}t\right)\mathbf{i} - 18t\,\mathbf{j}\right]$

Or either term correct

M1

$\mathbf{F} = -16\pi\sin\left(\frac{\pi}{3}t\right)\mathbf{i} - 72t\,\mathbf{j}$

A1

2

(ii) When $t = 3$, $\mathbf{F} = 4 \times [-4\pi\sin(\pi)\,\mathbf{i} - 54\mathbf{j}]$

$= -216\mathbf{j}$

B1

Magnitude is 216

ft finding magnitude of their F

B1ft

2

(c) $\mathbf{r} = \int \mathbf{v}\,dt$

either term correct

M1

$= \dfrac{36}{\pi}\sin\left(\dfrac{\pi}{3}t\right)$ **i** – 3t^3 **j** + **c**

No need for **c** *(otherwise CAO)*

Condone $\dfrac{12}{\left(\pi/3\right)}$

A1

When t = 3, **r** = 4**i** – 2**j**

M1

→ –81**j** + **c** = 4**i** – 2**j**

c = 4**i** + 79**j**

A1

r = $\left\{\dfrac{36}{\pi}\sin\left(\dfrac{\pi}{3}t\right)+4\right\}$ **i** + {79 – 3t^3}**j**

CAO

A1

5

[11]

Q5.

(a) $F^2 = 70^2 + 40^2 - 2 \times 40 \times 70\cos 150°$

M1: Use of cosine rule with an obtuse angle and a – sign.

M1

A1: Correct expression.

A1

$F = \sqrt{11350}$

dM1: Taking square root of a value > 6500. May be implied by final answer.

dM1

F = 107

A1: Correct resultant to 3sf or more.

Accept AWRT 106 or 107.

A1

4

OR by components,

$70 + 40\cos 30°(= 104.64)$

M1: Finding two perpendicular components of the resultant, with same force (usually the 40 N force) resolved in both expressions.
Allow consistent sin / cos confusion.

A1: Both components correct.
(Note that resolving parallel and perpendicular to the 40 N force gives components of 100.6 and 35)

(M1A1)

$40\sin 30°$ (= 20)

dM1: Finding the magnitude of the resultant.

(dM1)

$F = \sqrt{104.64^2 + 20^2}$ = 107 N

A1: Correct resultant to 3sf or more.
Accept AWRT 106 or 107.

(A1)

(4)

(b) $\dfrac{\sin\alpha}{40} = \dfrac{\sin 150°}{106.54}$

M1: Use of sine rule with 150°, their answer to part (a) and 40 or 70.
A1: Correct equation with AWRT 106 or 107.

M1A1

$\alpha = 10.8°$

A1: Correct angle. Accept 10.9°.

A1

3

OR

$\tan\alpha = \dfrac{20}{104.64}$

M1: Use of tan with 20 and AWRT 104 or 105.

A1: Expression for tan α in the form

$\tan\alpha = \dfrac{20}{\text{AWRT } 104 \text{ or } 105}$.
Could be implied by their final answer.

(M1A1)

$\alpha = 10.8°$

A1: Correct angle. Accept 10.9°.

(A1)

(3)

OR

$\sin \alpha = \frac{20}{106.53}$

M1: Use of sin with 20 or AWRT 104 or 105 in the numerator and their answer to (a) as the denominator.

A1: Expression for sin α in the form

$\sin \alpha = \frac{20}{\text{AWRT } 106 \text{ or } 107}$.

(M1A1)

$\alpha = 10.8°$

A1: Correct angle. Accept 10.9°

(A1)

(3)

OR

$\cos \alpha = \frac{104.64}{106.53}$

M1: Use of cos with 20 or AWRT 104 or 105 in the numerator and their answer to (a) as the denominator.

A1: Expression for cos α in the form

$\cos \alpha = \frac{\text{AWRT } 104 \text{ or } 105}{\text{AWRT } 106 \text{ or } 107}$.

(M1A1)

$\alpha = 10.8°$

A1: Correct angle. Accept 10.9° or 10.7°.

Apply ISW if 180° – α is seen after finding α.

(A1)

(3)

[7]

Q6.

(a) $8 = \frac{1}{2} \times 9.8t^2$

M1: Equation based on the vertical motion, with u = 0, s = ± 8 and a = ± 9.8.
A1: Correct equation.

M1A1

$t = \sqrt{\frac{16}{9.8}} = 1.28\ s$

A1: Correct time. Allow 1.27 or AWRT 1.28.

A1

3

(b) $V \times \sqrt{\frac{16}{9.8}} = 20$

M1: Using 20 = speed × time.
A1: Correct equation.

M1A1

$V = 20\sqrt{\frac{9.8}{16}} = 15.7\ \text{m s}^{-1}$

A1: Correct speed. Accept 15.6 or $7\sqrt{5}$ or AWRT 15.6 or AWRT 15.7.

A1

3

(c) $v_y = 9.8 \times \sqrt{\frac{16}{9.8}}$ (= 12.52)

M1: Finding vertical component of velocity, with u = 0, a = ±9.8 and their time from part (a).
A1: Correct expression for velocity.

M1A1

$v = \sqrt{15.65^2 + 12.52^2} = 20.0\ \text{m s}^{-1}$

dM1: Finding the magnitude (with addition).

dM1

A1: Correct speed.
Accept 20 or 20.1 or AWRT 20.0.

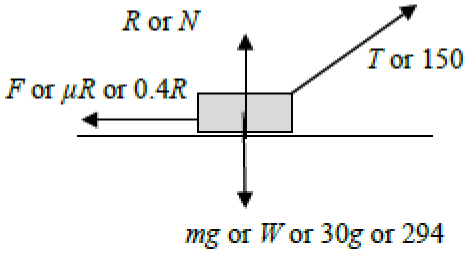
A1

4

[10]

Q7.

(a) (i)



B2: Correct diagram with exactly four forces showing arrow heads and labelled.
B1: Diagram with one error or omission.
B0: Diagram with 2 or more errors or omissions.

If components are also shown and they use a different style, eg dashed lines, they can be ignored.

If both components are shown in the same style as other forces, this counts as two errors.

Note; Do not accept 30kg for the weight.

B1

B1

2

(ii) $R + 150 \sin 20° = 30 \times 9.8$

$(R =)\ 30 \times 9.8 - 150 \sin 20°$

$= 242.69\ldots\ldots$

M1: Resolving vertically to obtain a three term equation, with R, 150 sin or cos(20° or 70°) and 30g oe.

A1: Correct equation. Allow g instead of 9.8.

M1A1

$= 243$ N (to 3sf)

A1: AG Correct final answer having seen either 2nd or 3rd or both line of solution.

A1

3

(iii) $(F =)\ 0.4 \times 242.7 = 97.1$ N

M1: Use of $F = \mu R$ or $F \le \mu R$.
A1: Correct final answer without an inequality.
Accept 97.2.

M1A1

2

(iv) $30a = 150 \cos 20° - 97.08$

M1: Three term equation of motion with 30a, 150 sin or cos(20 or 70°) and their friction from (a)(iii). Condone incorrect signs.
A1: Correct equation.

M1A1

$$a = \frac{150\cos 20° - 97.08}{30} = 1.46 \text{ m s}^{-2}$$

dM1: Solving for a.

dM1

A1: Correct acceleration. Accept 1.45 or 1.47 or AWRT 1.46.

A1

4

(b) $R = 30 \times 9.8 - T\sin 20°$

B1: Correct normal reaction in terms of T.

B1

$F = 0.4\,(30 \times 9.8 - T\sin 20°)$

B1: Correct friction in terms of T

B1

$T\cos 20° = 0.4\,(30 \times 9.8 - T\sin 20°)$

M1: Resolving tension horizontally and equating to F, provided that F is in terms of T.
A1: Correct equation.

M1A1

$$T = \frac{0.4 \times 30 \times 9.8}{\cos 20° + 0.4\sin 20°} = 109 \text{ N}$$

A1: Correct tension. AWRT 109.

A1

5

(c) The same

B1: The same.

Use of g = 9.81 gives acceptable final answers.

B1

1

[17]

Q8.

(a) $x = ut\cos\theta$

M1

$t = \dfrac{x}{u\cos\theta}$

A1

$y = -\dfrac{1}{2}gt^2 + ut\sin\theta$

Condone + g for M1

M1

$y = -\dfrac{1}{2}gt^2 + ut\sin\theta$

A1

$y = -\dfrac{1}{2}g\left(\dfrac{x}{u\cos\theta}\right)^2 + u\left(\dfrac{x}{u\cos\theta}\right)\sin\theta$

Elimination of t, condone + g for m1

m1

$y = -\dfrac{gx^2}{2u^2\cos^2\theta} + \dfrac{x\sin\theta}{\cos\theta}$

$y = -\dfrac{gx^2}{2u^2}(1 + \tan^2\theta) + x\tan\theta$

OE in terms of x, u, g, $\tan\theta$

A1

6

(b) (i) $0.5 = -\dfrac{9.8(5)^2}{2(8)^2}(1 + \tan^2\theta) + 5\tan\theta$

Correctly substituting for x, y, u and g into their equation of trajectory

M1

All correct, condone decimal approximation.

A1

$245 \tan^2 \theta - 640 \tan \theta + 309 = 0$

OE exact quadratic in tan θ

A1

$\tan \theta = \dfrac{640 \pm \sqrt{(-640)^2 - 4(245)(309)}}{2(245)}$

PI by the values of tan θ

m1

$\tan \theta = 1.973(004), \quad 0.6392(41)$

$\theta = 63.12°, \ 32.58°$

$\theta = 63.1°, 32.6°$

AG Must see the above or more accurate values

A1 **5**

(ii) $\dot{y} = -9.8\left(\dfrac{5}{8\cos 63.1°}\right) + 8 \sin 63.1°$ OE

Condone +9.8 for M1.

M1

$(\dot{y} = -6.4035)$

$\dot{x} = 8 \cos 63.1°$

$(\dot{x} = 3.6195)$

M1

$\tan^{-1} \dfrac{6.4(035)}{3.6(195)}$ (= 61°) OE

PI by correct angle in a statement

m1

Direction: 61° to the horizontal
or 29° to the vertical

Have to see "horizontal" or "vertical" or diagram

A1 **4**

<u>**Alternative:**</u>

$\frac{dy}{dx} = -\frac{2gx}{2u^2}(1 + \tan^2\theta) + \tan\theta$

(M1)

$= -\frac{2\times 9.8\times 5}{2\times 8^2}(1 + \tan^2 63.1°) + \tan 63.1°$

$= -1.7692$

(A1)

$\tan^{-1}(-1.7692) = -60.52368°$

(m1)

Direction: 61° to the horizontal
or 29° to the vertical

(A1)

(4)

(c) The ball is a particle, or
No air resistance, or
The ball does not spin

B1

1

[16]

Q9.

(a) (i) $10^2 = 4^2 + 2 \times a \times 50$

M1: Use of a constant acceleration equation to find a, with v and u substituted correctly.
For example $4^2=10^2 + 100a$ scores M0A0A0.
A1: Correct constant acceleration equation.

M1A1

$a = \frac{100-16}{100} = 0.84 \text{ ms}^{-2}$

A1: Correct a.

A1

Note if t found first award M1 for use of
$v = u + at$ or $s = ut + \frac{1}{2}at^2$.

3

(ii) $50 = \frac{1}{2}(4 + 10)t$

M1: Use of a constant acceleration equation to find t.

A1F: Correct constant acceleration equation with their acceleration from (a)(i) seen.

M1A1

$t = \frac{50}{7} = 7.14$ s

A1: Correct t. Accept $\frac{50}{7}$ or $7\frac{1}{7}$ or AWRT 7.14.

A1

OR

$10 = 4 + 0.84t$

If t has been found in part (a) the working does not have to be repeated, but value of t must be stated.

(M1A1F)

$t = \frac{6}{0.84} = 7.14$ s

Do not follow through incorrect values of a.

(A1)

OR

$50 = 4t + \frac{1}{2} \times 0.84t^2$

(M1A1F)

$0.42t^2 + 4t - 50 = 0$

$t = 7.14$ (or $t = -16.6$)

(A1)

3

(b) $70 \times 0.84 = 58.8$ N

M1: Use of $F = ma$ with $m = 70$ and their a from (a)(i).

A1F: Correct F. Follow through their value of a from part (a)(i).

M1A1F

2

(c) (i) $58.8 = 70 \times 9.8 \sin \alpha$

$\sin\alpha = \dfrac{58.8}{70 \times 9.8} = 0.08571$

M1: Resolving parallel to the slope must see 70g or mg OE with sin α or cos α and their answer to part (b).

A1F: Correct equation. Follow through their answer to part (b) provided sin α < 1

M1A1F

$\alpha = 4.92°$

A1F: Correct angle. Follow through their answer to part (b). Accept 4.91° provided sin α < 1.

A1F

3

(ii) $70 \times 9.8 \sin\alpha - 30 = 58.8$

$\sin\alpha = 0.12945$

M1: Three term equation of motion. Must see 70g or mg OE with sin α or cos α.

A1F: Correct equation. Follow through their answer to part (b) provided sin α < 1

M1A1F

$\alpha = 7.44°$

A1F: Correct angle. Follow through their answer to part (b) provided sin α < 1.

Accept 7.43°.

Accept 7.41° from 0.129.

A1F

3

(d) The air resistance force will increase (vary or change) with speed.

B1: Correct statement.

B1

1

[15]

Q10.

(a) $\tan\alpha = \dfrac{6}{10}$

M1: Using tan with 10 and 5 or 6, OR sin or cos with $\sqrt{136}$ and 6 or 10, OR sin or cos with $\sqrt{125}$ and 5 or 10.

Note: $\sin\alpha = \frac{6}{\sqrt{136}}$ *= and* $\cos\alpha = \frac{10}{\sqrt{136}}$

M1

$\alpha = 31.0°$

A1: Correct angle. Accept 30.9° or AWRT 31°

A1

2

(b) $8 \sin \alpha t + 4.9\, t^2 = 6$

*M1: equation for the vertical motion containing ±6 or ±5, ± 4.9*t^2 *and ±8sin*α *or ±8cos*α*, where* α *has a value related to their answer to part (a)*
(May be a negative angle).

M1

A1F: Correct terms.

A1F: Correct signs and terms
Follow through angle from part (a).

A1FA1F

$4.9t^2 + 4.116t - 6 = 0$

A1: Correct equation rearranged equal to zero, but may be implied by subsequent working.

A1

$t = 0.76359$ or $t = -1.60$ s

dM1: Attempting to solve their quadratic equation. Only award method mark if method seen or correct answers obtained or –0.764 with +1.60.

dM1

$t = 0.764$

A1: Correct solution obtained. Accept 0.763 or AWRT 0.764.

A1

OR

$v = \sqrt{(8\sin 31.0°)^2 + 2\times 9.8\times 6} = 11.60$

M1: Use a constant acceleration equation $v^2 = u^2 + 2as$ *to find* v*.*

(M1)

A1F: Correct equation.
A1F: Correct v.

(A1FA1F)

$11.60 = 8 \sin 31° + 9.8t$

dM1: Use of $v = u + at$ to find t

(dM1)

$t = \frac{11.60 - 8\sin 31°}{9.8} = 0.763$

A1: Correct equation.

(A1)

A1: Correct t (0.763)

(A1)

6

(c) $d = 10 - 8 \cos \alpha \times 0.764$

M1: Finding a horizontal distance using 8cosα or 8sinα multiplied by their time from part (b).
dM1: For subtracting their distance from 10.

M1dM1

$= 10 - 5.238$

A1: Seeing AWRT 5.24 or 5.23 from 0.763.

A1

$= 4.76$ m

A1: Correct final answer. Accept AWRT 4.76. Accept 4.77 from use of 0.763.

A1

4

[12]

Q11.

(a) $20 \cos\theta = 10$

M1: Resolving horizontally. Accept sin θ or cos θ with the 20.

$\cos \theta = \frac{1}{2}$

A1: Correct equation.

M1A1

$\theta = 60°$

A1: Correct angle.

Accept $\frac{\pi}{3}$ *or 1.05 (radians).*

Allow 59.9 or better if they find W first

A1

3

(b) $(W =)20 \sin 60°$

$= 17.3$ N

Or

$(W =)\sqrt{20^2 - 10^2} = 17.3$ N

M1: Resolving vertically. Accept sin θ or cos θ with the 20, where θ is their answer to part (a) or 90 minus their answer to part (a).

A1: Correct weight CSO

or

M1: Correct use of Pythagoras eg $10^2 + W^2 = 20^2$

A1: Correct weight CSO

Accept $10\sqrt{3}$ *or AWRT 17.3*

M1A1F

2

(c) $m = \dfrac{20\sin 60°}{9.8} = 1.77$ kg

M1: Their answer to part (b) divided by 9.8.

A1F: Correct mass. Follow through their answer to part (b).

Accept 1.76 or 1.8.

Accept 2 sig figs in follow through.

Note: Using g = 9.81 gives the answer 1.77, also accept 1.76.

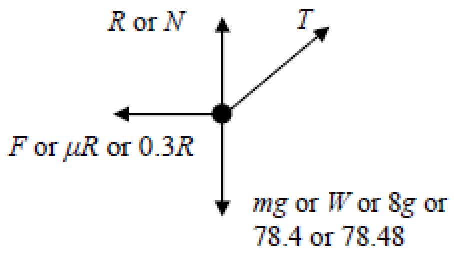
M1A1F

2

[7]

Q12.

(a)



*B1: Diagram with **exactly four** forces showing arrow heads and labelled. If components are also shown and they use a different style, eg dashed lines, they can be ignored.*

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 8 kg for the weight.

Note Accept μR or 0.3R for F.

B1

1

(b) R + 40 sin θ = 8 × 9.8

M1: Resolving vertically to obtain a three term equation, with R, T sin or cos(30° or 60°) and 8g oe.

A1: Correct equation

M1A1

R = 78.4 – 40sin θ

A1: Correct expression for R.

Accept (R=)8g – T sin30°

Note if using g = 9.81 accept

R = 78.48 – 0.5T or R = 78.5 – 0.5T

A1

3

(c) 8a = 40 cos θ – 0.3(78.4 – 40 sin θ)

M1: Use of the friction inequality with R from part (b)

A1: Correct equation.

M1A1

8a = – 23.52 + 40 cos θ + 12 sin θ

M1: Use of equation of motion with 40 sin θ or 40 cosθ and 8a and a friction term.

M1

p = – 2.94

q = 8

r = 1.5

A1: One correct value.

A1

A1: All three values correct.

A1

5

[9]

Q13.

(a) $22.4 \sin \theta - 2 \times 9.8 = 0$

M1: Use of $v = u + at$ vertically with $u = 22.4\sin\theta$, $v = 0$, $t = 2$ and $a = \pm 9.8$.

A1: Correct equation. (May be in terms of g or contain 9.81..

M1A1

$\sin \theta = \frac{19.6}{22.4} = \frac{7}{8} = 0.875$

A1: Must see either

$22.4 \sin \theta = 19.6$ or $\frac{19.6}{22.4}$

A1

OR

M1: Use of $s = ut + \frac{1}{2}at^2$ with $u = 22.4\sin\theta$, $s = 0$, $t = 4$ and $a = \pm 9.8$.

A1: Correct equation.

A1: must see $89.6 \sin \theta = 78.4$ or $\frac{78.4}{89.6}$ OE

3

(b) $h_{MAX} = 22.4 \times \frac{7}{8} \times 2 - \frac{1}{2} \times 9.8 \times 2^2$

M1: Using a constant acceleration equation to find height, with $t = 2$, $u = 22.4 \sin \theta$ or 19.6 and $a = \pm 9.8$.

A1: Correct equation.

M1A1

$= 19.6$ m

A1: Correct height. AWRT 19.6

Note using $g = 9.81$ gives 19.6, also accept 19.5.

Note: other constant acceleration equations will lead to the same result

A1

3

(c) $\cos\theta = \frac{\sqrt{15}}{8} = 0.4841$ or $\theta = 61.04°$

B1: Correct value for cos θ (accept 0.484) or θ (accept 61.0° or 61° or 1.06 or 1.065 or 1.07 radians). Can be implied.

B1

$AB = 22.4 \times \frac{\sqrt{15}}{8} \times 4 = 43.4$ m

M1: Calculation for range with value for cosθ and with t = 4.

A1F: Correct distance. Follow through incorrect θ. Accept AWRT 43.4 or 43.3 or 43.2.

Do not accept 43.

M1A1F

3

(d) $22.4 \times \frac{7}{8}t - 4.9t^2 = 5$

M1: Use of $s = ut + \frac{1}{2}at^2$ with correct terms, but not necessarily signs.

M1

$4.9t^2 - 19.6t + 5 = 0$

A1: Correct equation.

A1

$t = 0.274$ or $t = 3.726$

dM1: Solving their quadratic.

dM1

A1: At least one correct solution. Allow 0.27 or 0.28 and 3.72 or 3.73

A1

Time = 3.726 – 0.274 = 3.45 seconds

A1: Correct difference. Accept 3.46.

A1

Note: there are other methods which will lead to the correct time:
M1dM1A1 for a constant acceleration equation that gives a time or times from which the final answer can be obtained

A1 Correct time or times
A1 Correct final answer

5

[14]

Q14.

(a) $20 \cos \theta = 10$

M1: Resolving horizontally. Accept sin θ or cos θ with the 20.
A1: Correct equation.

M1A1

$\cos \theta = \frac{1}{2}$

$\theta = 60°$

A1: Correct angle.
Accept $\frac{\pi}{3}$ or 1.05 (radians).
Allow 59.9 or better if they find W first

A1

3

(b) $(W =)\ 20 \sin 60°$

M1: Resolving vertically. Accept sin θ or cos θ with the 20, where θ is their answer to part (a) or 90 minus their answer to part (a).

M1

$= 17.3$ N

A1: Correct weight CSO

A1

Or

$(W =)\sqrt{20^2 - 10^2} = 17.3$ N

M1: Correct use of Pythagoras eg $10^2 + W^2 = 20^2$

(M1)

A1: Correct weight CSO
Accept $10\sqrt{3}$ or AWRT 17.3

(A1)

2

(c) $m = \dfrac{20\sin 60°}{9.8}$

M1: Their answer to part (b) divided by 9.8.

M1

= 1.77 kg

A1F: Correct mass. Follow through their answer to part (b).

Accept 1.76 or 1.8.

Accept 2 sig figs in follow through.

Note: Using g = 9.81 gives the answer 1.77, also accept 1.76.

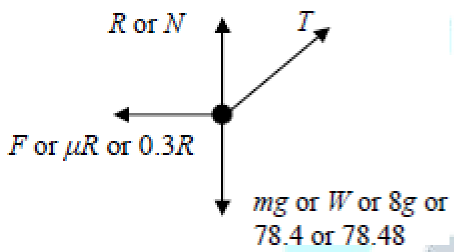
A1F

2

[7]

Q15.

(a)



*B1: Diagram with **exactly four** forces showing arrow heads and labelled. If components are also shown and they use a different style, eg dashed lines, they can be ignored.*

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 8 kg for the weight.

Note Accept μR or 0.3R for F.

B1

1

(b) $R + T\sin 30° = 8 \times 9.8$

M1: Resolving vertically to obtain a three term equation, with R, T sin or cos(30° or 60°) and 8 g oe.

A1: Correct equation

M1A1

$(R =)\ 78.4 - T\sin 30°$

$(R =)\ 78.4 - 0.5\,T$

A1: Correct expression for R.

Accept (R =) $8g - T \sin 30°$

Note if using $g = 9.81$ accept
$R = 78.48 - 0.5T$ or $R = 78.5 - 0.5T$

A1

3

(c) $T \cos 30° - F = 8 \times 0.05$

M1: Horizontal equation of motion with F, T sin or cos (30° or 60°) and 8×0.05 oe.

A1: Correct equation.

M1A1

$F = 0.3(78.4 - \sin 30°)$

M1: Using $F = 0.3R$ with their R from part (b), provided it includes a term in T.

A1: Correct expression for friction.

M1A1

$T \cos 30° - 0.3(78.4 - T \sin 30°) = 0.4$

$$T = \frac{23.52 + 0.4}{\cos 30° + 0.3 \sin 30°} = 23.5\text{N}$$

dM1: Solving for T. Must see (cos 30° ± 0.3sin 30°) or similar in the denominator. (Dependent on both previous M marks.)

A1: Correct T. Accept 23.6 or AWRT 23.5

dM1A1

6

Or

$T \cos 30° - F = 8 \times 0.05$

M1: Horizontal equation of motion with F, T sin or cos(30° or 60°) and 8 × 0.05 oe.

A1: Correct equation.

(M1A1)

$T \cos 30° - 0.3R = 8 \times 0.05$

M1: Using $F = 0.3R$

A1: Two correct equations involving only T and R.

(M1A1)

$R + T\sin 30° = 8 \times 9.8$

solving simultaneously gives

$T = 23.5$

dM1: Solving for T.

A1: Correct T. Accept 23.6 or AWRT 23.5

Note using g = 9.81 gives 23.6, also accept 23.5.

(dM1A1)

[10]

Q16.

(a) $22.4 \sin\theta - 2 \times 9.8 = 0$

M1: Use of $v = u + at$ vertically with
$u = 22.4\sin\theta$, $v = 0$, $t = 2$ and $a = \pm 9.8$.

A1: Correct equation. (May be in terms of g or contain 9.81..

M1A1

$\sin\theta = \frac{19.6}{22.4} = \frac{7}{8} = 0.875$

A1: Must see either

$22.4 \sin\theta = 19.6$ or $\frac{19.6}{22.4}$

AG

A1

Or

$0 = 22.4 \sin\theta \times 4 - \frac{1}{2} \times 9.8 \times 4^2$

M1: Use of $s = ut + \frac{1}{2}at^2$ with $u = 22.4\sin\theta$, $s = 0$, $t = 4$ and $a = \pm 9.8$.

A1: Correct equation.

(M1A1)

$\sin\theta = \frac{4.9 \times 16}{22.4 \times 4} = 0.875$

A1: must see $89.6 \sin\theta = 78.4$ or $\frac{78.4}{89.6}$ OE

(A1)

(b) $h_{MAX} = 22.4 \times \sin\theta \times 2 - \frac{1}{2} \times 9.8 \times 2^2$

M1: Using a constant acceleration equation to find height, with t = 2, u = 22.4 sin θ or 19.6 and a = ±9.8.

A1: Correct equation.

M1A1

= 19.6 m

A1: Correct height. AWRT 19.6

Note using g = 9.81 gives 19.6, also accept 19.5.

Note: other constant acceleration equations will lead to the same result

A1

Or

$0^2 = (22.4 \times \sin\theta)^2 + 2 \times (-9.8)h_{MAX}$

(M1A1)

h_{MAX} = 19.6m

(A1)

3

(c) $\cos\theta = \frac{\sqrt{15}}{8}$ = 0.4841 or θ = 61.04°

B1: Correct value for cosθ (accept 0.484) or θ (accept 61.0° or 61° or 1.06 or 1.065 or 1.07 radians). Can be implied.

B1

$AB = 22.4 \times \frac{\sqrt{15}}{8} \times 4$ = 43.4 m

M1: Calculation for range with value for cosθ and with t = 4.

A1F: Correct distance. Follow through incorrect θ. Accept AWRT 43.4 or 43.3 or 43.2.

Do not accept 43.

M1A1F

3

(d) $22.4 \times (\sin\theta)t - 4.9t^2 = 5$

M1: Use of $s = ut + \frac{1}{2}at^2$ with correct *terms, but not necessarily signs.*

M1

$4.9t^2 - 19.6t + 5 = 0$

A1: Correct equation.

A1

$t = 0.274$ or $t = 3.726$

dM1: Solving their quadratic.

dM1

A1: At least one correct solution. Allow 0.27 or 0.28 and 3.72 or 3.73

A1

Time = 3.726 – 0.274 = 3.45 seconds

A1: Correct difference. Accept 3.46.

A1

Note: there are other methods which will lead to the correct time:

M1dM1A1 for a constant acceleration equation that gives a time or times from which the final answer can be obtained

A1 Correct time or times

A1 Correct final answer

5

(e) $v_{MIN} = 22.4 \times \cos\theta$

M1: Finding horizontal component with candidate's value for cos θ. Do not award if combined with a non-zero vertical component.

M1

$= 10.8 \text{ m s}^{-1}$

A1: Correct speed. Accept 10.9 or 10.85.

A1

Or

$v_{\min} = \frac{43.4}{4} = 10.9$ to 3sf

M1: range divided by time of flight

(M1)

A1: Correct speed. Accept 10.9 or 10.85.

(A1) 2

[16]

Q17.

(a) $x = ut\cos\alpha$

M1

$t = \dfrac{x}{u\cos\alpha}$

A1

$y = -\dfrac{1}{2}gt^2 + ut\sin\alpha$

Must have correct signs

M1

$y = -\dfrac{1}{2}g\left(\dfrac{x}{u\cos\alpha}\right)^2 + u\left(\dfrac{x}{u\cos\alpha}\right)\sin\alpha$

M1

$y = -\dfrac{gx^2}{2u^2\cos^2\alpha} + \dfrac{x\sin\alpha}{\cos\alpha}$

$y = -\dfrac{gx^2}{2u^2}(1 + \tan^2\alpha) + x\tan\alpha$

A1

$k = -\dfrac{10(2k)^2}{2u^2}(1 + \tan^2\alpha) + 2k\tan\alpha$

M1

$u^2 = -20k(1 + \tan^2\alpha) + 2u^2\tan\alpha$

$20k\tan^2\alpha - 2u^2\tan\alpha + u^2 + 20k = 0$

AG

A1 7

(b) Pass through $P \Rightarrow$ Discriminant ≥ 0

$(-2u^2)^2 - 4(20k)(u^2 + 20k) \geq 0$

OE must be seen

M1A1

$4u^4 - 80ku^2 - 1600k^2 \geq 0$

$u^4 - 20ku^2 - 400k^2 \geq 0$

AG

A1

3

[10]

Q18.

(a) $T = 2 \quad 9.8 = 19.6$ N

M1: Equating tension and weight.

A1: Correct tension CAO

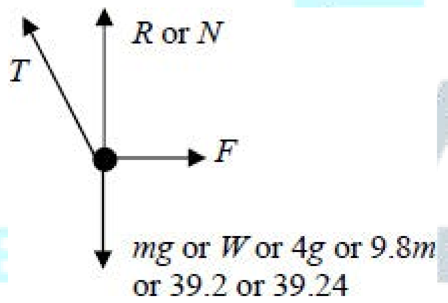
Accept $2g$

Accept 19.62 from $g = 9.81$

M1A1

2

(b)



B1: R, F (not μR) and mg correct

B1

B1: T correct, must be in roughly correct direction.

If more than four forces shown, do not award more than one mark.

Note all forces must be shown as arrows and have labels.

Note some candidates may draw the force diagram in the section with the question.

Components can be ignored if shown in a different notation eg dashed arrows.

B1

2

(c) $T \cos 30° + R = 4 \quad 9.8$

M1: Resolving vertically to form a three term equation. (May be implied)

M1

(R =) 39.2 – 19.6cos30°
= 39.2 – 16.9741...

A1 Correct expression for R or equation for R.
Must see 19.6cos30 or equivalent (eg 2$g$sin60)

A1

= 22.2259....
= 22.2 N (to 3sf) AG

A1: Correct force. Must see intermediate working, for example third or fourth line of working in solution opposite.

Example: 19.6sin 30° – R = 4 9.8 scores M1A0A0.

Use of g = 9.81 still gives 22.2 N as the final answer.

A1

3

(d) T cos 60° = F

M1: Resolving horizontally

M1

F = 19.6cos60° = 9.8

A1: Correct expression for friction

A1

19.6cos60° ≤ μ(39.2 – 19.6cos30°)

dM1: Use of $F = \mu R$ or $F \le \mu R$ (do not allow $F \ge \mu R$)

dM1

$$\mu \ge \frac{19.6cos60°}{39.2 - 19.6cos30°}$$

$\mu \ge 0.441$

A1: Final answer of μ = 0.441 or $\mu \ge 0.441$ from correct working

Use of g = 9.81 still gives 0.441 as the final answer.

A1

4

[11]

Q19.

(a) $12\sin30°t - 4.9t^2 = -0.5$

$4.9t^2 - 12\sin30°t - 0.5 = 0$

M1: Three term equation for vertical motion, with ±g, ±0.5 (or ±1 and ±1.5) and 12sin 30°t or 12cos30°t.

A1: Correct terms. (one must be equivalent to ±0.5)

A1: Correct signs.

M1A1A1

$t = 1.30281...$ *or* $-0.078323...$

dM1: Solving the quadratic to find t.

Must see use of quadratic equation formula or can be implied by seeing 1.303 or 1.302 or similar.

dM1

$t = 1.30$ seconds (to 3sf) AG

A1: Correct time from correct working. Must see more than 3 significant figures in candidate's working before the final answer or two correct solutions to the quadratic (eg 1.3 and –0.08).

Accept 1.3

A1

OR
time up = 0.6122
time down = 0.6122 + 0.0783 = 0.6905
total time = 0.6122 + 0.6905 = 1.30 (to 3sf)

M1: Adding time up to time down having used a quadratic.

A1: 0.6122

dM1: Finding time down with a quadratic

A1: 0.6905

A1: Correct answer

Accept 1.3

(M1A1dM1A1A1)

OR
$-6.767 = 12\sin 30° - gt$

M1:Forms an equation to find t having found v first

A1: Correct terms

A1: Correct signs

(M1A1A1)

$t = \dfrac{12\sin 30° + 6.767}{g}$ = 1.30281 = 1.30 (to 3sf)

dM1: Solving for t

A1: Correct time from correct working. Must see more than 3 significant figures in candidate's working before the final answer.

Accept 1.3

(dM1A1)

5

(b) 12cos30° 1.303 = 13.5 m

M1: Finding horizontal displacement using 1.30 (or better) and 12cos30°.

Do not allow 12sin30°.

A1: Correct distance. AWRT 13.5.

M1A1

2

(c) v_y = 12sin30° – 9.8 1.3028 (= –6.767)

M1: Finding vertical component of velocity or velocity squared at impact.

Must include 12sin30° or 12cos30 and ±g

A1: Correct expression for vertical component. May have 1.3 or 1.30 instead of 1.3028. (Accept +6.767 or similar)

M1A1

$v = \sqrt{(12\cos 30°)^2 + (-6.767)^2} = 12.4 \text{ m s}^{-1}$

dM1: Finding speed from two components. May use 6.74.

A1: Correct speed. Allow 12.3 or AWRT 12.4.

Note using g = 9.81 still gives 12.4.

dM1A1

4

(d) $\tan\theta = \dfrac{6.767}{12\cos 30°}$

M1: Trigonometric equation to find angle.

Can only be those shown opposite or described below. For tan, fraction can be inverted. For sin, 10.4 can be used instead of 6.767. For cos, 6.767 can be used instead of 10.4. Can use their values from part (c) (eg 6.74 or 6.77).

M1

θ = 33.1°

A1F: Correct angle. Accept AWRT 33°.

A1F

OR

$\sin\theta = \dfrac{6.767}{12.4}$

θ = 33.1°

OR

$\cos\theta = \dfrac{10.4}{12.4}$

θ = 33.1°

Follow though vertical component or final speed from part (c).

2

(e) The weight is the only force acting.
OR
No air resistance.

B1: Appropriate assumption.

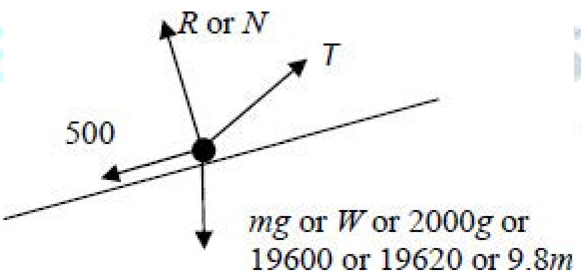
B1

1

[14]

Q20.

(a)



B1: R, 500 and mg correct

B1

B1: Tension in roughly correct direction.

If more than four forces shown, do not award more than one mark.

Note all forces must be shown as arrows and have labels.

Note some candidates may draw the force diagram in the section with the question.

Components can be ignored if shown in a different notation eg dashed arrows.

B1

2

(b) 2000 0.6 = T cos12° – 500 – 2000 9.8sin5°

M1: Resolving parallel to the slope to obtain a four term equation of motion. The weight and tension terms must be resolved.

A1: Correct terms.

A1: Correct signs.

M1A1A1

$T = \dfrac{1200 + 500 + 19600 sin5°}{cos12°}$

$\left(= \dfrac{3408.25}{cos12°}\right)$

(= 3484.4)

dM1: Solving for T.

dM1

= 3480 (to 3sf) AG

A1: Correct tension. AWRT 3480.

Allow AWRT 3490 from use of g = 9.81.

A1

5

[7]

Q21.

(a) 1000 = V × 4

M1: Equation for horizontal motion to find V.
Must not contain g. Could contain cos0° or equivalent.

M1

V = 250 ms^{-1}

A1: Correct V.

A1

2

(b) $(h =)\dfrac{1}{2} \times 9.8 \times 4^2$

M1: Vertical equation to find height with u = 0 *and a = ±9.8.*

M1

= 78.4 meters to 3sf

A1: Correct height. Accept –78.4

A1

2

(c) $(v_y =)9.8 \times 4 = 39.2 \text{ ms}^{-1}$

M1: Calculation of vertical component of velocity with u = 0 and a = ±9.8.

A1: Correct vertical component.

M1A1

or

$$(v_y =)\sqrt{2 \times 9.8 \times 78.4} = 39.2 \text{ ms}^{-1}$$

$$(v_y =)\sqrt{250^2 + 39.2^2} = 253 \text{ ms}^{-1}$$

dM1: Calculation of speed.

A1: Correct speed.

dM1A1

4

(d) $$\tan\alpha = \frac{39.2}{250}\left(\text{or } \tan\alpha = \frac{250}{39.2}\right)$$

M1: Using tan to find angle with opposite and adjacent sides. Can be inverted as shown in brackets.

A1F: Correct trig expression.

M1A1F

α = 8.91°

A1: Correct angle.

A1

OR

$$\sin\alpha = \frac{39.2}{253}\left(\text{or } \sin\alpha = \frac{250}{253}\right)$$

M1: Using sin to find angle with hypotenuse and one other side. Can be changed as shown in brackets.

A1F: Correct trig expression.

(M1A1F)

α = 8.91°

A1: Correct angle.

(A1)

OR

$$\cos\alpha = \frac{250}{253(.055)} \left(\text{or } \cos\alpha = \frac{39.2}{253} \right)$$

M1: Using cos to find angle with hypotenuse and one other side. Can be changed as shown in brackets.

A1F: Correct trig expression.

(M1A1F)

α = 8.91°

A1: Correct angle. Accept 8.83° from this method.

(A1)

Note: Accept 8.98° from 253.1

Accept negative angles

Note: FT value of V from (a) and speed from (c) if needed. Do not FT 39.2 from (c) in place of 253.

Note: Accept energy methods if used correctly in part (c).

3

[11]

Q22.

(a) P cos80° – Q cos80° = 250a

M1: Horizontal equation of motion in the form
P cos80° ± Q cos80° = 250a or
P sin80° ± Q sin80° = 250a

A1: Correct horizontal equation.

M1A1

P sin80° + Q sin80° = 250g

B1: Correct vertical equation.

Note: the above marks could be awarded for a correct vector equation.

B1

$$P - Q = \frac{250a}{\cos 80°}$$

$$P + Q = \frac{250g}{\sin 80°}$$ **AG**

$$2P = \frac{250a}{\cos 80°} + \frac{250g}{\sin 80°}$$

dM1: Solving for P with an attempt to eliminate Q.

dM1

$$P = 125\left(\frac{a}{\cos 80°} + \frac{g}{\sin 80°}\right)$$

A1: Correct result from correct working.
Must see an expression for 2P or 2P sin80°cos80°

A1

5

(b) P cos80° = 250a

M1: Using Q = 0 into correct original equation(s) or resolving without Q.

M1

P sin80° = 250g

$$\frac{1}{\tan 80°} = \frac{a}{g}$$

dM1: Eliminating P

dM1

$$a = \frac{g}{\tan 80°} = 1.73$$

A1: Correct a.
Note: use of P = ± Q scores M0dM0A0
Note: use of P = 0 can lead to ± 1.73 but scores M0dM0A0 unless fully justified by a symmetry argument.

A1

3

[8]

Q23.

(a) x = 40cos$\theta.t$

M1

$$y = -\frac{1}{2}(10)t^2 + 40\ \sin\theta.t$$

M1 A1

$$y = -\frac{1}{2}(10)\left(\frac{x}{40\ \cos\theta}\right)^2 + 40\ \sin\theta.\left(\frac{x}{40\ \cos\theta}\right)$$

Dependent on both M1s

m1

$$y = -\frac{x^2}{320\cos^2\theta} + x\tan\theta$$

320 y = –x^2(1 + tan² θ) + 320x tan θ

m1

$x^2 \tan^2 \theta - 320x \tan \theta + (x^2 + 320y) = 0$

Answer Given (Condone missing brackets)

A1

6

(b) (i) $150^2 \tan^2 \theta - 320(150)\tan \theta + (150^2 + 320 \times 8) = 0$

M1

$1125 \tan^2 \theta - 2400 \tan \theta + 1253 = 0$

Correct quadratic

A1

$$\tan \theta = \frac{2400 \pm \sqrt{2400^2 - 4(1125)(1253)}}{2(1125)}$$

m1

$\tan \theta = 1.22$, 0.912

PI

A1F

$\theta = 50.7°$, $42.4°$

A1F

5

(ii) $\theta = 42.4°$

For the smaller angle

B1F

$t = \dfrac{150}{40\cos \theta}$ and $\cos 42.4 > \cos 50.7$

OE

E1

2

[13]

Q24.

(a) $5 = \dfrac{1}{2} \times 9.8t^2$

M1: Equation based on vertical motion with no velocity component, with ± 5 and ± 9.8

M1

A1: Correct equation

A1

$t = \sqrt{\dfrac{5}{4.9}} = 1.01$ s **AG**

A1: Correct time from correct working.

Must see square root or $t^2 = 1.02$ OE

Note: If $g = 9.81$ is used for the first time deduct one mark. Should still get 1.01 seconds.

A1

3

(b) $15 = V \times \sqrt{\dfrac{5}{4.9}}$

M1: Using distance = speed × time OE

M1

$V = 15\sqrt{\dfrac{4.9}{5}} = 14.8$

A1: Correct speed.
Accept AWRT 14.8 or 14.9.

Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 14.9 m s^{-1} from $g = 9.81$.

A1

2

(c) $v_v = \pm 9.8 \times \sqrt{\dfrac{5}{4.9}}$ $(= \pm 9.899)$

M1: Calculating vertical component of velocity.
A1: Correct value. Accept 9.9 or similar

M1A1

or

$v_v = \sqrt{2 \times 9.8 \times 5} = 9.899$

$v = \sqrt{9.899^2 + 14.8^2} = 17.8$ m s^{-1}

dM1: Finding magnitude (with addition not subtraction of squares inside the square root).

dM1

A1: Correct speed.
Accept AWRT 17.8 or AWRT 17.9.

Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 17.9 m s^{-1} from $g = 9.81$

A1F

4

(d) $\tan\alpha = \frac{9.899}{14.8}$ or $\frac{14.8}{9.899}$

$\alpha = 34°$

M1: Use of one of trig equations shown.

M1

A1F: Anything which rounds to 34° or 56°

A1F

A1F: 34° CAO (33° scores M1A1A0)

A1F

3

$\sin\alpha = \frac{9.899}{17.8}$ or $\frac{14.8}{17.8}$

$\alpha = 34°$

Only follow through if all method marks in (b) and (c) have been awarded (except the dM if tan used).

(M1)(A1F)(A1F)

$\cos\alpha = \frac{14.8}{17.8}$ or $\frac{9.899}{17.8}$

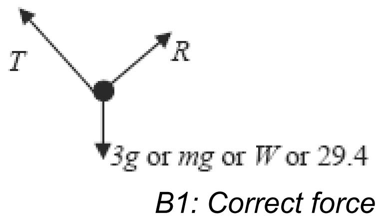
$\alpha = 34°$

(M1)(A1F)(A1F)

[12]

Q25.

(a)



B1: Correct force diagram with arrows and sensible labels.

If R is shown as vertical award B0.

If F is included, award B0

Accept a reflection of the diagram in a vertical line.

Ignore components if shown with a different notation, eg dotted lines.

B1

1

(b) (R =) $3g$ cos 60

M1: Resolving perpendicular to the slope. Must see cos 60° or sin 30° or cos 30° or sin 60° and $3g$ or 29.4.

NOTE: $\frac{3g}{2} = 14.7$ *or equivalent without the use of a trig term scores M0.*

M1

(R =) 14.7 **AG**

A1: Correct value from correct working.
NOTE: If candidates use g = 9.81, deduct one mark here. If candidates obtain 14.7 from 14.715 they will have used g = 9.81.
Note: "R =" does not need to be seen.

A1

2

(c) (T =) $3g$ sin 60°

M1: Resolving parallel to the slope. Must see cos 60° or sin 30° or cos 30° or sin 60° and $3g$ or 29.4.

M1

(T =) 25.5

A1: Correct value. AWRT 25.5 or truncation to 25.4.
NOTE: If candidates use g = 9.81 again, do not penalise. Use of g = 9.81 gives 25.5 for the tension.
Note: "T =" does not need to be seen.

A1

2

[5]

Q26.

(a) $5 = \frac{1}{2} \times 9.8t^2$

M1: Equation based on vertical motion with no velocity component, with ± 5 and ± 9.8

M1

A1: Correct equation

A1

$t = \sqrt{\frac{5}{4.9}} = 1.01 \text{ s}$ **AG**

A1: Correct time from correct working.
Must see square root or $t^2 = 1.02$ OE
Note: If $g = 9.81$ is used for the first time deduct one mark. Should still get 1.01 seconds.

A1

3

(b) $15 = V \times \sqrt{\frac{5}{4.9}}$

M1: Using distance = speed × time OE

M1

$V = 15\sqrt{\frac{4.9}{5}} = 14.8$

A1: Correct speed.
Accept AWRT 14.8 or 14.9.
Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 14.9 m s^{-1} from $g = 9.81$.

A1

2

(c) $v_v = \pm 9.8 \times \sqrt{\frac{5}{4.9}} (= \pm 9.899)$

M1: Calculating vertical component of velocity.
A1: Correct value. Accept 9.9 or similar

M1A1

or

$v_v = \sqrt{2 \times 9.8 \times 5} = 9.899$

$v = \sqrt{9.899^2 + 14.8^2} = 17.8 \text{ m s}^{-1}$

dM1: Finding magnitude (with addition not subtraction of squares inside the square root).

dM1

A1: Correct speed.
Accept AWRT 17.8 or AWRT 17.9.

A1F

Note: If g = 9.81 is used for the first time deduct one mark. Should get 17.9 m s^{-1} from g = 9.81

4

(d) $\tan\alpha = \frac{9.899}{14.8}$ or $\frac{14.8}{9.899}$

$\alpha = 34°$

M1: Use of one of trig equations shown.

M1

A1F: Anything which rounds to 34° or 56°

A1F

A1F: 34° CAO (33° scores M1A1A0)

A1F

Only follow through if all method marks in (b) and (c) have been awarded (except the dM if tan used).

3

$\sin\alpha = \frac{9.899}{17.8}$ or $\frac{14.8}{17.8}$

$\alpha = 34°$

(M1)(A1F)(A1F)

$\cos\alpha = \frac{14.8}{17.8}$ or $\frac{9.899}{17.8}$

$\alpha = 34°$

(M1)(A1F)(A1F)

(e) Particle

B1: Particle assumption

B1

Experiences no air resistance **or** no wind **or** only gravity **or** no other forces acting **or** no spin.

B1: Other assumption.
Ignore any other assumptions.

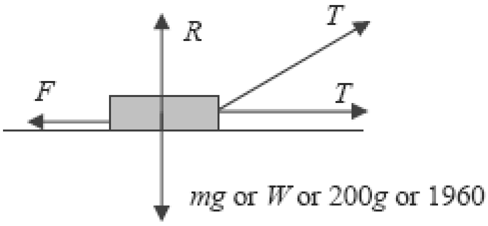
B1

2

[14]

Q27.

(a)



B1: F, R and mg (or equivalent) with arrows and labels.

B1

*B1: Two **equal** tension forces with arrows and labels.*

B1

Ignore components if shown with a different notation, eg dotted lines.

2

(b) $R + T\sin 20° = 1960$
OR
$R + T\sin 20° = 200g$

M1: Resolving vertically with three terms. Must include sin 20° or cos 20° or sin 70° or cos 70° with T and 200g or 1960.
A1: Correct equation.

M1A1

(R =) 1960 – $T\sin 20°$
OR
(R =) 200g – $T\sin 20°$

A1: Correct expression for the normal reaction.
Note: If g = 9.81 is used for the first time deduct one mark. Should get 1962 instead of 1960

A1

3

(c) $T\cos 20° + T - F = 200 \times 0.3$

M1: Four term equation of motion. Must include sin 20° or cos 20° or sin 70° or cos 70° with T and a second T term with no trig.
A1: Correct equation

M1A1

$T\cos 20° + T - 0.4\,(1960 - T\sin 20°) = 200 \times 0.3$

M1: Use of friction law with their expression for R, *provided that R has two terms. Note that this mark does not depend on any previous marks.*

Example:
If Candidate gives 1960 as answer to part (b), then: F = 0.4 × 1960 = 784 scores M0 here

M1

$$T = \frac{60 + 784}{\cos 20° + 1 + 0.4 \sin 20°} = 406$$

dM1: Solving for T.

dM1

Note: This mark requires both of the previous M marks.
A1: Correct tension.
Accept AWFW 406 to 407.
Note: If g = 9.81 is used should get 407 instead of 406.

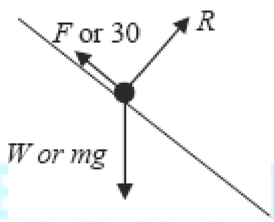
A1

5

[10]

Q28.

(a)



B1: Correct force diagram with arrows and labels.

B1

1

(b) $30 = mg \sin 42°$

M1: Two term equation by resolving parallel to the slope. Must include g.
A1: Correct equation.

M1A1

$$m = \frac{30}{9.8 \sin 42°} = 4.57$$

A1: Correct value for m.

A1

3

(c) (R =) mg cos 42° = 33.3 N

M1: Resolving perpendicular to the slope to find R. Must include g.
A1: Correct value.

M1A1F

2

(d) 30 = μR

$$\mu = \frac{30}{R} = 0.900$$

M1: Use of $F = \mu R$

M1

M1: Solving for μ.

M1

A1F: Correct value for μ. Accept 0.9.

A1F

3

Use of g = 9.81

(b) 4.57	*B1*
(c) 33.3	*B1*
(b) 0.900	*B1*

[9]

Q29.

(a) 14.7 sin α – 9.8t (= 0)

M1: Equation for vertical velocity being zero at highest point. Must have sin α with ± 9.8.
A1: Correct equation.

M1A1

$$t = \frac{14.7\sin\alpha}{9.8} = \frac{3\sin\alpha}{2}$$ **AG**

A1: Correct result from correct working.

A1

OR

$$14.7\sin\alpha T - 4.9T^2 (= 0)$$

$$T = \frac{14.7\sin\alpha}{4.9} = 3\sin\alpha$$

$$t = \frac{3\sin\alpha}{2}$$

All marks awarded for last line, from correct working.

(M1)(A1)(A1) **3**

(b) (i) $7 = 14.7 \sin\alpha\left(\frac{3\sin\alpha}{2}\right) - 4.9\left(\frac{3\sin\alpha}{2}\right)^3$

M1: Expression including vertical displacement at height 7, using expression from part (a) and with ± g or equivalent.

M1

A1: Correct expression.

A1

$7 = 11.025\sin^2\alpha$

dM1: Simplified expression with $\sin^2\alpha$.

dM1

dM1: Finding an angle. Must have previous dM1 mark.

dM1

$\alpha = \sin^{-1}\left(\sqrt{\frac{7}{11.025}}\right) = 52.8°$

A1: Correct angle.
Accept 52.7°, 52.9°.

A1

OR

$0^2 = (14.7\sin\alpha)^2 + 2\times(-9.8)\times 7$

(M1)(A1)

$\sin^2\alpha = \frac{2\times 9.8\times 7}{14.7^2}$

(dM1)

$\alpha = 52.8°$

(dM1)(A1) **5**

(ii) $OA = 14.7\cos 52.8° \times 3\sin 52.8°$

B1: Use of 3sin α with their α.
M1: Finding horizontal displacement.

including 14.7 cos α with 3 sin α or $\frac{3\sin\alpha}{2}$.

B1M1

OA = 21.2 m

A1: Correct distance. Accept 21.3 m.

A1

3

Use of g = 9.81:
(a) M1A1A0
(b) (i) 52.8° or 52.9° M1A1dM1dM1A1
(b) (ii) 21.2 B1M1A1

[11]

Q30.

(a) $14.7\sin\alpha - 9.8t (= 0)$

M1: Equation for vertical velocity being zero at highest point. Must have sin α with ± 9.8.
A1: Correct equation.

M1A1

$$t = \frac{14.7\sin\alpha}{9.8} = \frac{3\sin\alpha}{2}$$ **AG**

A1: Correct result from correct working.

A1

OR

$14.7\sin\alpha T - 4.9T^2 (= 0)$

$T = \frac{14.7\sin\alpha}{4.9} = 3\sin\alpha$

$t = \frac{3\sin\alpha}{2}$

All marks awarded for last line, from correct working.

(M1)(A1)(A1)

3

(b) (i) $$7 = 14.7\sin\alpha\left(\frac{3\sin\alpha}{2}\right) - 4.9\left(\frac{3\sin\alpha}{2}\right)^2$$

M1: Expression including vertical displacement at height 7, using expression from part (a) and with ±g or equivalent.

M1

A1: Correct expression.

A1

$7 = 11.025\sin^2\alpha$

dM1: Simplified expression with $\sin^2\alpha$.

dM1

dM1: Finding an angle. Must have previous dM1 mark.

dM1

$\alpha = \sin^{-1}\left(\sqrt{\dfrac{7}{11.025}}\right) = 52.8°$

A1: Correct angle.
Accept 52.7°, 52.9°.

A1

OR

$0^2 = (14.7\sin\alpha)^2 + 2\times(-9.8)\times 7$

(M1)(A1)

$\sin^2\alpha = \dfrac{2\times 9.8\times 7}{14.7^2}$

(dM1)

$\alpha = 52.8°$

(dM1)(A1)

5

(ii) OA = 14.7 cos 52.8° × 3 sin 52.8°

B1: Use of 3sin α with their α.
M1: Finding horizontal displacement.

including 14.7 cos α with 3 sin α or $\dfrac{3\sin\alpha}{2}$.

B1M1

OA = 21.2 m

A1: Correct distance. Accept 21.3 m.

A1

3

(c) Ball is a particle/No spin.

B1: Particle assumption.

B1

No air resistance/No wind/Constant acceleration of 9.8/Only force is weight.

B1: Air resistance assumption.

B1

2

[13]

Use of g = 9.81:
(a) M1A1A0
(b) (i) 52.8° or 52.9° M1A1dM1dM1A1
(b) (ii) 21.2 B1M1A1

Q31.

(a) (i) $x = 80\cos\theta.t$

B1

$$t = \frac{x}{80\cos\theta}$$

B1

$$y = 80\sin\theta.t - \frac{1}{2}gt^2$$

B1

$$y = 80\sin\theta\frac{x}{80\cos\theta} - \frac{1}{2}g\left(\frac{x}{80\cos\theta}\right)^2$$

M1

$$y = x\tan\theta - \frac{gx^2}{12800}(1+\tan^2\theta)$$

Answer given

A1

5

(ii) $$-20 = 400\tan\theta - \frac{9.8\times 400^2}{12800}(1+\tan^2\theta)$$

Condone + 20

M1

$$122.5\tan^2\theta - 400\tan\theta + 102.5 = 0$$
$$49\tan^2\theta - 160\tan\theta + 41 = 0$$

Answer given

A1

2

(b) (i) $$\tan\theta = \frac{160 \pm \sqrt{25600 - 4(49)(41)}}{2\times 49}$$

M1

= 2.9850, 0.2803

PI

A1

$$\theta = 71.5°, 15.7°$$

A1F

3

(ii) For the shortest time
$400 = 80\cos 15.7°.t$

M1

$t = 5.19$

A1F

2

(c) • The projectile is a particle
• The air resistance is negligible

E1

1

[13]

Q32.

(a) $0^2 = (30\sin 35°)^2 + 2 \times (-9.8)s$

M1: Equation to fine the max height, with = 0

M1

A1: Correct equation

A1

M1: Solving for the height

M1

$$s = \frac{(30 \sin 35°)^2}{2 \times 9.8} = 15.1 \text{ m}$$

A1: Correct height

A1

4

(b) $28^2 = (30 \cos 35°)^2 + v_y^2$

M1: Equation to find vertical component

M1

A1: Correct equation

A1

$$v_y = \sqrt{28^2 - (30 \cos 35°)^2}$$

M1: Solving equation

M1

= 13.4198 = 13.4 m s^{-1} **AG**

A1: Correct speed from correct working.

A1

4

(c) $\tan \theta = \dfrac{13.4}{30 \cos 35°}$

M1: Expression to find the angle.

M1

$\theta = 28.6°$

A1: Correct angle.

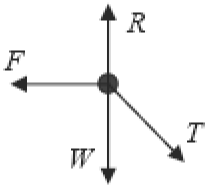
A1

2

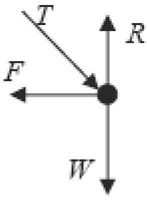
[10]

Q33.

(a)



OR



B1: Diagram with four forces showing arrow heads and labelled.
Allow mg or $8g$.
Allow T or 40 or other reasonable notation.
Allow μR.
Direction of friction must be to the left.
Any components must be shown in a different style.

B1

1

(b) $8g + 40 \sin 30°$ (= R)

M1: Expression for normal reaction, with mg or $8g$ and $40 \sin 30°$ or $40 \cos 30°$.
Allow incorrect signs.

M1

A1: Correct expression with correct signs.

A1

(R =) 98.4 N **AG**

A1: Correct value from correct working.
Use of g = 9.81 gives 98.5 N. Do not enalise if you have already done so arlier in the scripts.
Otherwise penalise by 1 mark.

A1

3

(c) $F = 40 \cos 30° = 34.6$ N

M1: Use of $40 \cos 30°$ or $40 \sin 30°$.
Award M0 if any extra terms

M1

A1: Correct value for friction. Don't need to see F.

A1

2

(d) $40 \cos 30° \le \mu \times 98.4$

M1: Use or $F \le \mu R$ (or $F = \mu R$). Must use R = 98.4 and a positive value for F.

M1

A1F: Correct inequality or equation Allow use of $F = \mu R$ throughout.

A1F

$$\mu \ge \frac{40\cos 30°}{98.4}$$
$$\mu \ge 0.352$$

A1F: Correct minimum value. For follow through must use R = 98.4 and their value for F from part (c). For example use of sin 30° in part (c) gives 0.203.

A1F

3

[9]

Q34.

If candidates have already used g = 9.81 do not penalise again on this question.

(a) $0^2 = (28\sin 50°)^2 + 2 \times (-9.8)s$

M1: Equation to find the max height, with v = 0, u = 28 sin 50° or u = 28 cos 50° and – 9.8 or $-g$

M1

A1: Correct equation

A1

$$s = \frac{(28\sin 50°)^2}{2 \times 9.8} = 23.5 \text{ m}$$

dM1: Solving for the height

dM1

A1: Correct height. Awrt 23.5
Note: If using a memorised formula, either 4 marks if final answer correct, 3 marks if substituted correctly but evaluated incorrectly, otherwise zero.

A1

OR
$0 = 28 \sin 50° - 9.8t$

M1: Equation to find time to the max height, with v = 0, u = 28 sin 50° or u = 28 cos 50° and – 9.8 or – g

(M1)

$t = \dfrac{28 \sin 50°}{9.8} = 2.1887$

A1: Correct time

(A1)

$s = 28 \sin 50° \times 2.1887 - 4.9 \times 2.1887^2 = 23.5$

dM1: Finding the height with their time and u = 28 sin 50° or u = 28 cos 50° and –4.9 or –g/2

(dM1)

A1: Correct height. Awrt 23.5

(A1)

4

(b) $2 = 28 \sin 50°\, t - 4.9t^2$

M1: Quadratic equation in t with a ± 2, u = 28 sin 50° or u = 28 cos 50° and –4.9 or –g/2.

M1

A1: Correct terms

A1

A1: Correct signs for equation

A1

$0 = 4.9t^2 - 28 \sin 50°\, t + 2$
$t = 0.0953$ or $t = 4.282$

dM1: Solving the quadratic equation

dM1

$t = 4.282 = 4.28$ s (to 3 sf) **AG**

A1: Correct larger time selected from two values.

A1

OR

M1: Calculation of two times, which sum or differ to give the time of flight.

(M1)

$0 = 28\sin 50° - 9.8t$

$t = \dfrac{28\sin 50°}{9.8} = 2.1887$

A1: Correct time by equation for zero vertical component of velocity or maximum height.

(A1)

OR

$23.5 = 28\sin 50° t - 4.9t^2$

$t = 2.1887$

$21.5 = 4.9t^2$

dM1: Correct expression for time to fall.

(dM1)

$t = \sqrt{\dfrac{21.5}{4.9}} = 2.0947$

A1: Correct time.

(A1)

2.1887 + 2.0947 = 4.2834 = 4.28 (to 3 sf) **AG**

A1: Correct time. Accept 4.29 if their answer rounds to 4.29.

(A1)

5

(c) $v_x = 28\cos 50° (= 18.00 \text{ m s}^{-1})$

B1: Horizontal component, need not be evaluated.

B1

$v_y = 28\sin 50° - 9.8 \times 4.282 = -20.51 \text{ m s}^{-1}$

M1: Equation for vertical component with 28 sin 50° (or 28 cos 50° if sin 50° used for horizontal component), −9.8 and awrt 4.28.

M1

A1: Correct vertical component. Awrt ± 20.5

A1

$v = \sqrt{18.00^2 + 20.51^2} = 27.3 \text{ m s}^{-1}$

dM1: Finding speed with a + sign inside the square root.

dM1

A1F: Correct speed. Awrt 27.3.

A1F

5

Intermediate values can be implied by final answer.

[14]

Q35.

(a) $h = \frac{1}{2} \times 9.8 \times 0.6^2 = 1.76$ m AG

M1: Equation to find h

M1

A1: Correct value from correct working

A1

2

(b) x = 18 × 0.6 = 10.8 m

M1: Calculating range

M1

A1: Correct value

A1

2

(c) $v_y = 9.8 \times 0.6 = 5.88 \text{ m s}^{-1}$

B1: Correct vertical

$v = \sqrt{5.88^2 + 18^2} = 18.9 \text{ m s}^{-1}$

M1: Calculating speed

M1

A1: Correct speed

A1

$$\theta = \tan^{-1}\left(\frac{5.88}{18}\right) = 18.1°$$

M1: Calculating angle using tan

M1

A1: Correct angle

A1

18.9 m s^{-1} at 18.1° below the horizontal.

B1: States below horizontal

B1

6

[10]